Anche se la comunicazione è importante è solo un lato della medaglia, come i processi cooperano e sono sincronizzati è l'altro lato delle medaglia.
In un sistema centralizzato il tempo non è ambiguo. Quando un processo vuole conoscere il tempo fa una semplice chiamata al sistema operativo. Se un processo A chiede il tempo e dopo poco il processo B chiede anch'esso il tempo, il valore che B riceverà sarà certamente più alto o al limite uguale al valore ricevuto da A, non sarà di certo più basso.

Time in distributed systems

- In a DS, processes are executed on network-connected nodes that cooperate to complete a computation and communicate via message exchange
- Observations:
  - Algorithms require synchronization
  - Processes running on different nodes in the DS must have a common notion of time to perform synchronized actions based on time
  - Algorithms require event ordering, meaning determining which event occurred before another
  - In DS, time is important but problematic

All'interno di un sistema distribuito, i nodi devono essere d'accordo dal punto di vista temporale ma anche perché si eseguono algoritmi che richiedono l'ordinamento degli eventi del sistema.

Time in distributed systems

- In a centralized system, it is possible to determine the order in which events occurred come già visto
  - Shared memory and single clock
- In a DS, it is impossible to have a single physical clock common to all processes
- Yet, the global computation can be viewed as a total order of events, by considering the time at which events were generated
- For many distributed algorithms, it is crucial to determine this time or, at the very least, establish the ordering of events: **how?**

Time in distributed systems: solutions

- Solution 1: synchronizing **physical clocks**
  - A physical clock counts the number of seconds
  - Each node in the DS adjusts its physical clock to be consistent with the clocks of other nodes or with a reference clock

i clock fisici però non sono precisi per loro natura. Dietro c'è un grosso discorso che parte dagli orologi atomici al cesio 133 per arrivare al sistema Coordinated Universal Time (UTC) (vedi libro Tanenbaum)

- Solution 2: synchronizing **logical clocks**
  - A logical clock counts the number of events
  - In a DS, physical clock synchronization is not necessary; only the ordering of events is required (proposed by Leslie Lamport)

un logical clock altro non è che un numero intero

Rispetto al termine classico di nodi in un sistema distribuito, useremo il termine processo per indicare i partecipanti che si coordinano tra loro per portare a termine un algoritmo tramite scambio di messaggi.

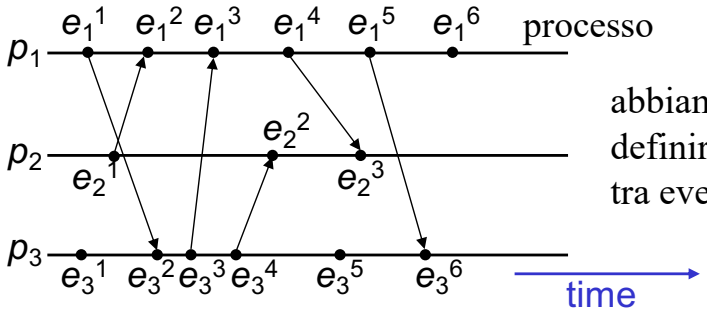
Model of distributed computation

- Components of DS: *N* processes and communication channels
- Each process p_i ($1 \le i \le N$) generates a sequence of events
  - e_i^k: *k*-th event generated by p_i
  - Internal events: state changes in the process
  - External events: send/receive messages
- Evolution of computation represented with space-time diagram

  $\rightarrow_i$: ordering relation between two events in p_i

  $e \rightarrow_i e'$ if and only if *e occurred before e'* in p_i ordinamento per eventi nello stesso processo

ovviamente ciascun processo conosce l'ordine dei propri eventi

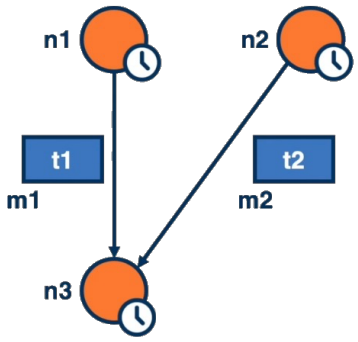


abbiamo come obiettivo definire l'ordinamento tra eventi di processi diversi

Timestamping

- Each process labels events with a timestamp
- Simple solution: each process labels events according to its own physical clock
- Does this work?
  - We can reconstruct the ordering of events on the same node
  - But what about the ordering of events on different nodes?
- In a distributed system, it is impossible to have a single physical clock shared by all processes

nella pratica il formato di un timestamp può essere in formato unix time



Synchronous vs. asynchronous DS

- Properties of a **synchronous DS** che devono essere soddisfatte!
  1. There are constraints on the execution speed of each process
     - The execution time of each step is limited, with both lower and upper bounds
  2. Each message transmitted on a communication channel is received within a limited time
  3. Each process has a physical clock with a known and limited clock drift rate from the real clock
     - Clock drift: deviation of clock from the true time over a period

esistono anche però, sistemi parzialmente sincroni

- In an **asynchronous DS**
  - *There are no constraints* on the execution speed of processes, the message transmission delay, or the clock drift rate

più flessibilità ma comunque serve ordine

Solutions to synchronize clocks

- First solution
  - Synchronize physical clocks of processes with some approximation using synchronization algorithms
  - Each process labels events with the value of its physical clock (which is synchronized with the other clocks with a certain degree of approximation)
  - Timestamping based on physical time (*physical clock*)

- Is it always possible to keep the approximation of physical clocks limited?
  - No, in an asynchronous DS
  - In an asynchronous DS, timestamping cannot be based on physical time; instead, it must be based on logical time (*logical clock*)

Physical clock

- At real-time instant t, the OS reads the time from the hardware clock $H_i(t)$ of the computer and generates the software clock $C_i(t) = aH_i(t) + b$ which approximately measures the physical time instant t for p_i b è l'istante iniziale mentre a è uno scalare che serve a convertire in ms
  - E.g., $C_i(t)$ is a 64-bit number that represents the number of nanoseconds elapsed from a reference instant until time t
  - In general, the clock is not perfectly accurate and may differ from t
  - If C_i behaves well enough, it can be used for event timestamping in p_i
- What should the clock resolution be in order to distinguish two events?
  - Smaller than the time difference ΔT between two relevant events la granularità dell'orologio è fondamentale per distinguere gli eventi

Physical clock in a DS

- In a SD physical clocks are different and can have different values
- *Skew*: instantaneous difference between the values of two clocks
- *Drift*: clocks measure time at different rates (due to physical variations), so over time they diverge from real time prese due macchine diverse, con il tempo i clock indicheranno tempi diversi
- *Drift rate*: difference per unit of time between a physical clock and an ideal clock
  - E.g., drift rate of 2 μsec/sec means that the clock increases by 1 sec + 2 msec every sec
  - Drift rate of standard quartz clocks: 10^{-6} s/s (~1 s in 11-12 days)
  - Drift rate of high-precision quartz clocks: 10^{-7} or 10^{-8} s/s

Universal Coordinated Time (UTC)

- International reference time standard
- Based on atomic time, occasionally corrected using astronomical time
  - 1 second is defined as the time is takes for a cesium-133 atom to complete 9,192,631,770 transitions
- Physical clocks using atomic oscillators are extremely accurate, with a drift rate as low as 10^{-13} s/s
- Atomic clock outputs are broadcast by terrestrial radio stations and satellites (e.g., GPS)
  - In Italy, Istituto Galileo Ferraris
- Nodes equipped with receivers can synchronize their clocks with these signals
  - Accuracy of signals from terrestrial radio stations: 1 to 10 ms
  - Accuracy of signals from satellites: 0.5 ms down to 50 ns

L'obiettivo degli algoritmi di sincronizzazione dei clock è di mantenere una deviazione tra i rispettivi clock tra due macchine qualunque in un sistema distribuito, all'intero di un limite detto precisione π :

$$\forall t, \forall p, q : |C_p(t) - C_q(t)| \leq \pi$$

Synchronizing physical clocks

- How to synchronize physical clocks with the atomic clock or with each other?
- **External** synchronization: the clocks C_i (i = 1, 2, … N) are synchronized with a time source S (UTC), such that, for a given real-time interval I

  $|S(t) - C_i(t)| \leq \alpha$ for $1 \leq i \leq N$ in the interval I

  – The clocks C_i have an **accuracy** of α, where $\alpha>0$

.

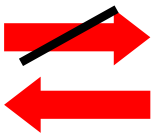
- **Internal** synchronization: two clocks C_i and C_j are synchronized with each other, so that

  $|C_i(t) - C_j(t)| \leq \pi$ for $1 \leq i,j \leq N$ in the interval I

  – The clocks C_i *and* C_j have a **precision** of π, where $\pi>0$

Synchronizing physical clocks

Internal synchronization



External synchronization

- Internally synchronized clocks are not necessarily synchronized externally
  - All clocks can collectively drift from an external source while still remaining synchronized with each other within a bound *D*

- If the set of processes is externally synchronized with accuracy of α, then it is also internally synchronized with a precision of 2α

dunque $\pi = 2\alpha$

Correctness of physical clocks

- A hardware clock *H is* correct if its drift rate is between $-\rho$ and $+\rho$ with $\rho > 0$
- If clock H is correct, the error made when measuring a real-time interval $[t, t']$ (with $t'>t$) is bounded:

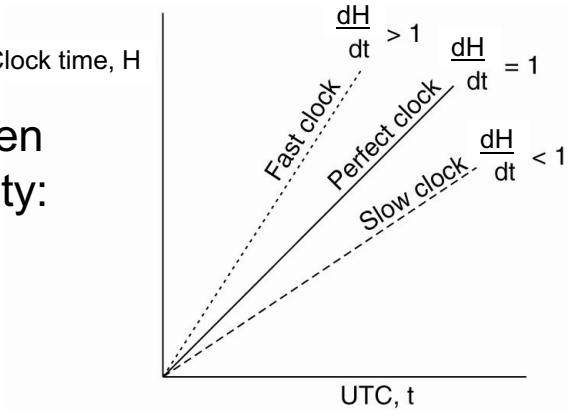
$$(1 - \rho)\,(t' - t) \leq H(t') - H(t) \leq (1 + \rho)\,(t' - t)$$

  - This avoids "jumps" in the clock value

- For a software clock C, it is often sufficient to require monotonicity:

$$t' > t \Rightarrow C(t') > C(t)$$

il tempo può soltanto crescere



When to synchronize physical clocks?

- Due to the drift rate, after a certain time, the clocks in a DS will become misaligned again → periodic synchronization is required to realign them
- Let's consider two clocks with the same maximum drift rate ρ from UTC
- Assume that after synchronization, the two clocks drift in opposite directions from UTC
  - After Δt from synchronization, they will have deviated by at most $2\rho\Delta t$
- To ensure that the difference between the two clocks remains within a desired tolerance δ, they must be resynchronized at least every $\delta/2\rho$

Internal synchronization in synchronous DS

- Internal synchronization algorithm between 2 processes in a synchronous DS
  - p_1 sends its local clock value t to p_2 via a message m, with transmission time T_{trasm}
  - p_2 receives m and sets its clock to $t + T_{trasm}$
    - T_{trasm} is unknown, but since the system is synchronous $T_{min} \leq T_{trasm} \leq T_{max}$
    - Let $u = (T_{max} - T_{min})$ be the transmission time uncertainty
  - Clock setting rule: if p_2 sets its clock to $t + (T_{max} + T_{min})/2$, the optimal lower bound on the clock skew is $u/2$
- Generalization: the algorithm can be extended to N processes
  - The optimal lower bound on the clock skew is $u(1 - 1/N)$
- Asynchronous DS
  - Message delays are unknown and unbounded: $T_{trasm} = T_{min} + x$, with $x \geq 0$ and unknown
  - Different physical clock synchronization algorithms are required

T_{trasm} non è noto a priori ma essendo sincrono il sistema abbiamo il lower e l'upper bound

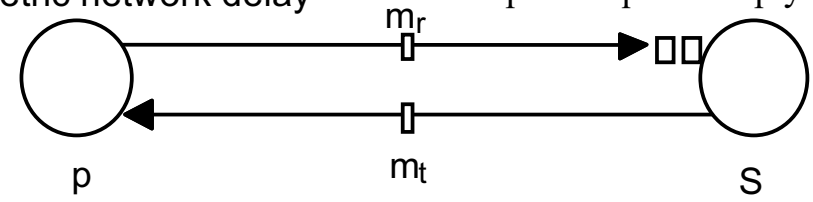
Physical synchronization via a time service

- A time service can provide the current time with a given precision
  - Equipped with a UTC receiver or a high-accuracy clock
- The group of processes that needs to synchronize uses a time service
  - Time service types: centralized or distributed
- Centralized time service
  - Request-driven: **Cristian's algorithm** (1989)
    - External synchronization
  - Broadcast-based: **Berkeley Unix algorithm** - Gusella & Zatti (1989)
    - Internal synchronization
- Distributed time service
  - **Network Time Protocol** (NTP) usato nella pratica oggi
    - External synchronization
    - Internet-scale

basato sulla stessa idea appena vista

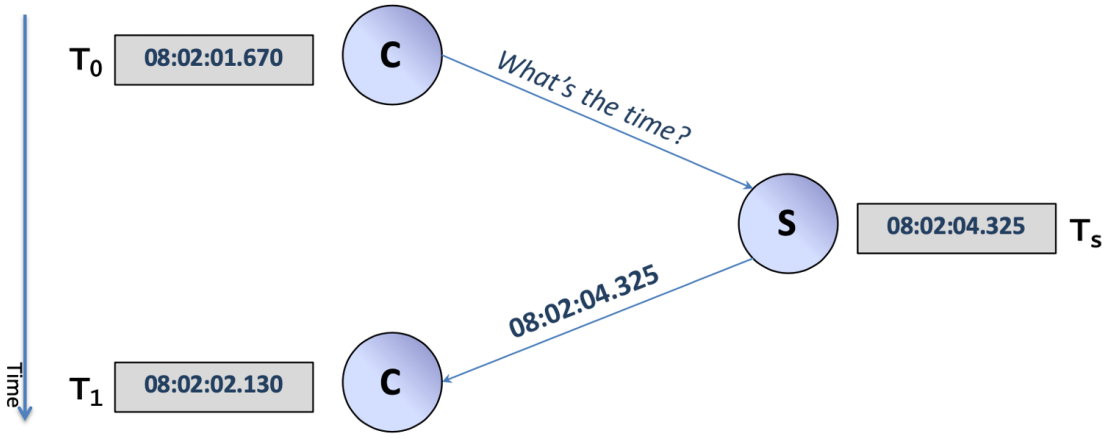
Cristian's algorithm

- A (passive) time server *S* receives the signal from a UTC source → external synchronization
- A process *p* requests the time by sending a request m_r and receives the time *t* in the reply m_t from *S*
- Process *p* sets its clock to $t + T_{round}/2$
  - T_{round} is the round-trip time measured by *p*, assume symmetric network delay ovvero i tempi di request e reply sono uguali



- Observations
  - A single time server may fail
    - Solution: use a group of synchronized time servers
  - Malicious time servers are not handled risolvibile tramite autenticazione
  - Reasonable accuracy only if T_{round} is small → suitable for low-latency LAN

Cristian's algorithm: example



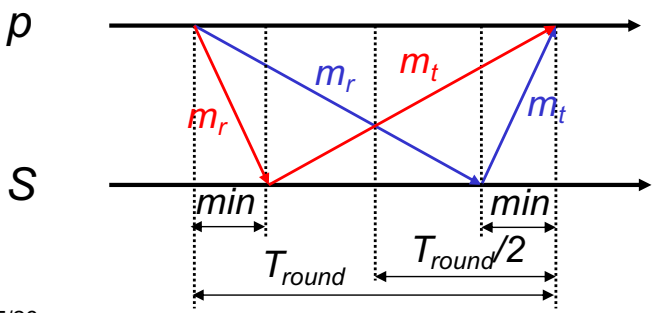
- $T_{round} = T_1 - T_0 = 460$ ms
- C sets its clock to $T_s + T_{round}/2$ = 08:02:04.325 + 230 ms = 08:02:04.555
- C clock increases by 2.425 s

Cristian's algorithm: accuracy

Case 1: *S* cannot insert *t* into m_t before *min* time units have elapsed since *p* sent m_r
- *min* is the minimum transmission time between *p* and *S*

Case 2: *S* cannot insert *t* into m_t after the instant when m_t arrives at *p* minus *min* tempo minimo di trasmissione tra client e server

- The time at *S* when m_t arrives at *p* lies in the interval $[t + min, t + T_{round} - min]$
  - Interval width: $T_{round} - 2\ min$
- Accuracy α: $\pm(T_{round}/2 - min)$



l'accuracy dipende da T_{round} dunque questo algoritmo non è adatto a reti sparse geograficamente

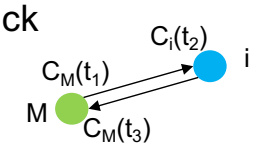
Berkeley algorithm

- Internal synchronization algorithm for a group of nodes in cui il master comunica il tempo ai worker
- The *master* node (*active* time server) broadcasts a request to all nodes, including itself, to get the clock values of the other nodes (*workers*)
- The master uses the RTTs to estimate the clocks of the workers
  - δ_i: difference between *M* clock and worker *i* clock (calculated similarly to Cristian's algorithm)

    $\delta_i = (C_M(t_1) + C_M(t_3))/2 - C_i(t_2)$

    $C_M(t_1)$ and $C_M(t_3)$: master's clock at sending and receiving

    $C_i(t_2)$: worker *i* clock when reply is sent

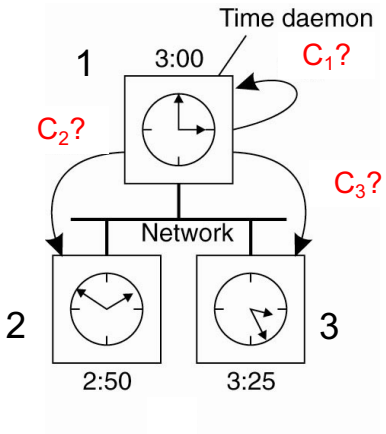


l'interazione tra master e worker è simile all'algoritmo di Cristian

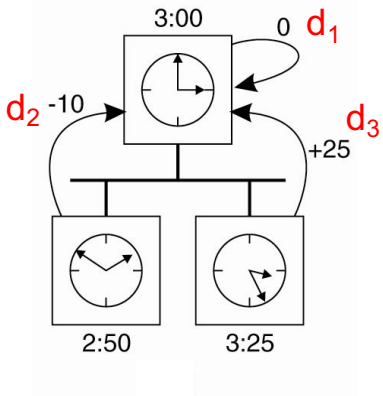
- Master calculates the average of of all δ_i (including its own) and sends correction value to workers
  - If backward adjustment needed: slow down clock

Berkeley algorithm: example

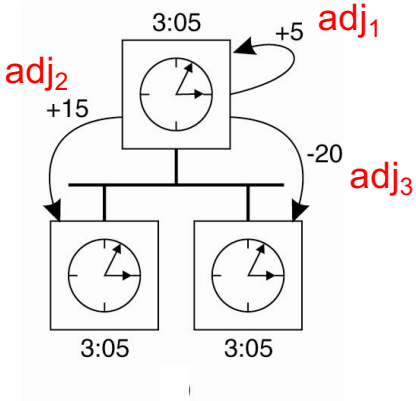
l'aggiustamento del tempo in avanti non porta problemi mentre il portare indietro il tempo viola il requisito della monoticità del tempo. In realtà il clock 3 rallenterà l'avanzata del tempo con interrupt, per colmare il divario. Il numero di interrupt è pari al tempo di solo down diviso il tempo di interrupt della CPU



Master broadcasts a request to all nodes



Master calculates the average of clock differences:
avg = (-10+25+0)/3 = 5



Master sends a correction value to each worker (adj_i = avg - d_i)

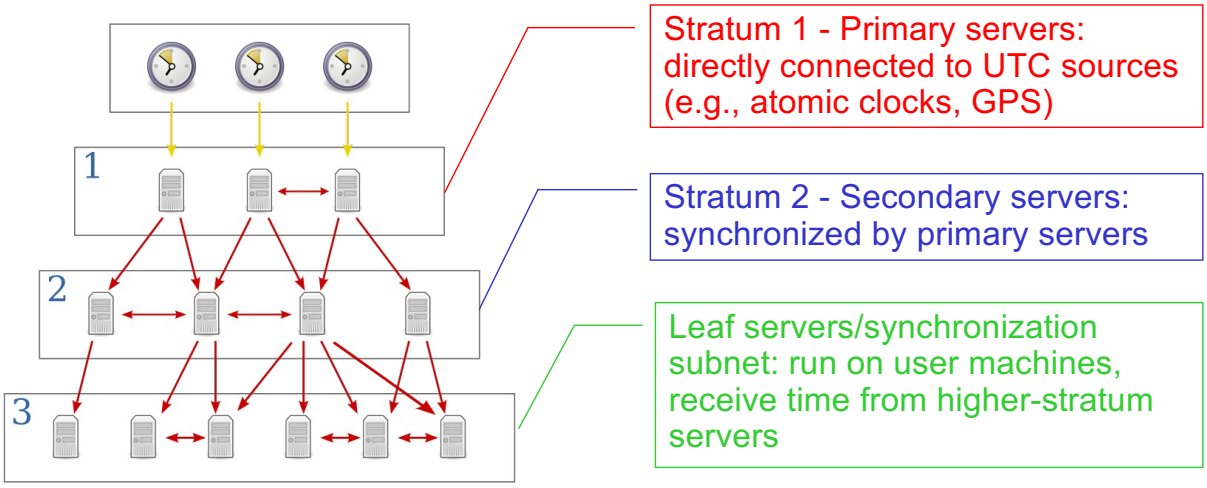
Berkeley algorithm: features

- Precision: depends on the maximum nominal RTT
  - The master ignores clock values associated with RTTs above this maximum
  - Outliers are discarded to improve accuracy

- Fault tolerance
  - Master failure: another node is elected as master using an election algorithm
  - Tolerance to arbitrary (Byzantine) behavior:
    - Workers sending incorrect clock values
    - The master only considers clock values that differ from each other by at most a specified threshold

Network Time Protocol (NTP)

- Time service for Internet (RFC 5905)
  - Provides external synchronization accurate wrt UTC
  - Configurable on multiple OSs
  - On GNU/Linux: `ntpd` daemon for automatic synchronization, `ntpdate` for manual synchronization
- Hierarchy of servers



da notare le interconnessioni esterne e interne
e come gli strati inferiori si sincronizzano con quelli superiori

NTP

- Architecture
  - Scalable and robust time service
  - Redundant time servers and network paths
  - Authentication of time sources to prevent

- NTP fault tolerance and reconfiguration
  - The synchronization subnet reconfigures automatically in case of failures
  - Primary server loses connection to UTC source: it becomes a secondary server
  - Secondary server loses connection to its primary (e.g., primary crashes): it can switch to another primary server

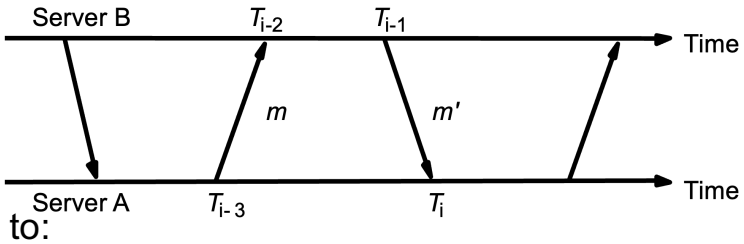
NTP: synchronization modes

1. Multicast mode
   - Server in a high-speed LAN multicasts its clock to others
   - Other servers set their clocks assuming a certain transmission delay
   - Accuracy: relatively low (no way to measure RTT)
2. Procedure call mode
   - Server responds to requests like in Cristian's algorithm non vedremo come
   - Accuracy: higher than multicast
   - Use case: when multicast is not available, makes sense for smaller networks
3. Symmetric mode
   - Server pairs exchange timing information
   - Accuracy: very high
   - Commonly used for higher-level hierarchical servers (Stratum 2 and above)

- All synchronization modes use UDP perché serve la velocità non fault tolerance

NTP: symmetric mode

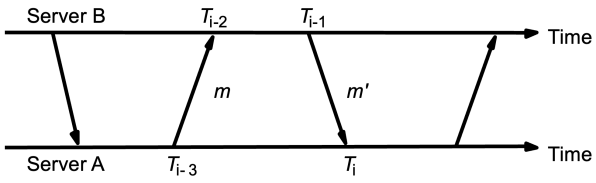
- Servers A and B exchange message pairs (*m*, *m'*) to improve synchronization accuracy
- Servers exchange messages with timestamps
  - A sends to B a message *m, which* contains:
    - Timestamp T_{i-3}: time when A sends *m*
  - B responds to A with message *m*', which contains:
    - Timestamp T_{i-2}: time when B receives *m*
    - Timestamp T_{i-1}: time when B sends *m'*
  - A records T_i when it receives *m'*
  - Time difference T_{i-1}-T_{i-2} may be non-negligible



- Timestamps allow servers to:
  - Estimate round-trip delay
  - Adjust their clocks more precisely

NTP: symmetric mode

- For each pair of messages *m* and *m'* exchanged between two servers, NTP estimates:
  - Offset o_i between the two clocks
  - Round-trip delay d_i, which is the total transmission time of *m* and *m'*



- Let's define:
  - *o*: real offset of clock B relative to clock A ($o = clock_B - clock_A$)
  - *t* and *t'*: actual transmission times of messages m and m', respectively
- For each pair of messages:

$$T_{i-2} = T_{i-3} + t + o \quad \text{and} \quad T_i = T_{i-1} + t' - o$$

$$\text{where } d_i = t + t' = T_{i-2} - T_{i-3} + T_i - T_{i-1}$$

- Subtracting the equation for T_i from the one for T_{i-2}, we solve for *o*:

$$o = [(T_{i-2} - T_{i-3}) + (T_{i-1} - T_i)]/2 + (t' - t)/2$$

Difference between the transmission times used to adjust the offset estimate

The estimated offset o_i is given by: $o_i = [(T_{i-2} - T_{i-3}) + (T_{i-1} - T_i)]/2$

The real offset *o* is: $o = o_i + (t' - t)/2$

NTP: symmetric mode

- The accuracy of the offset estimation is determined by the round-trip delay d_i
  - Since both t and t' are positive, the real offset o will lie within: $o_i - d_i/2 \leq o \leq o_i + d_i/2$
  - Thus, o_i is the estimated offset and its *accuracy* is $d_i/2$
- NTP servers apply a statistical filtering algorithm on the 8 most recent $<o_i, d_i>$ pairs, choosing as the estimate of o the o_j value corresponding to the minimum d_j
- They then apply a peer selection algorithm to possibly change the peer used for synchronization
  https://www.eecis.udel.edu/~mills/ntp/html/warp.html
- NTP accuracy:
  - 10 ms on the Internet
  - 1 ms on LAN

Perfect synchronization over networks is actually impossible

Google's TrueTime (TT)

- Distributed synchronized clock with bounded non-zero error
  - Designed by Google for Spanner, a global-scale, multi-version, distributed NewSQL database
  - Relies on a well-engineered tight clock synchronization available on all Google servers thanks to GPS and atomic clocks
  - Enables applications to generate monotonically increasing timestamps
  - *Cons*: TT requires special hardware and custom-build tight clock synchronization protocol, which is infeasible for many systems
    - Google also relies on its very high throughput, global fiber optic network linking its data centers

Time in asynchronous DS

- Physical clock synchronization algorithms
  - They (e.g., NTP) estimate transmission times to synchronize clocks between nodes
  - Accuracy depends on knowing the upper and lower bounds of transmission times
- Limitations in asynchronous DS
  - No constraints on transmission times → physical synchronization is used in asynchronous DS but with limited accuracy due to variable transmission times non si possono effettuare stime
  - We cannot use physical time to order events occurring on different nodes
- Logical time as a solution
  - What really matters is that processes agree on the order of events rather than the exact physical time at which events occurred
  - We use logical clocks (e.g., Lamport clocks) to maintain event ordering across nodes

Se due processi non interagiscono, non è necessario che siano sincronizzati perché la mancanza di sincronizzazione non sarebbe osservabile e non causerebbe alcun problema. Inoltre non tutti i processi devono concordare sul tempo, ma dovrebbero concordare sull'ordine in cui gli eventi occorrono.

Logical time

- Idea: order events based on two intuitive observations:
  1. Two events that occurred on the same process p_i happen exactly in the order observed by p_i
  2. When a message is sent from p_i to p_j, the send event happens before the receive event questo perché serve un tempo non nullo per ricevere un messaggio.
- Lamport (1978) and the **happened-before** relation
  - Leslie Lamport introduces the concept of the happened-before relation (also known as *causal ordering* or *precedence relation*)

  $\rightarrow_i$: ordering relation between two events that occurred on p_i

  $\rightarrow$: happened-before relation between any two events

L. Lamport, Time, Clocks and the Ordering of Events in Distributed Systems, 1978 https://lamport.azurewebsites.net/pubs/time-clocks.pdf

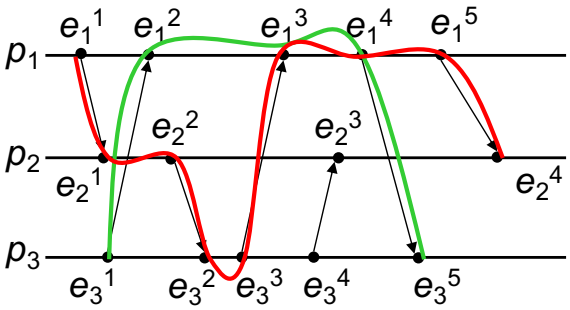
Happened-before relation

- Two events *e* and *e'* are in the happened-before relation (denoted by $e \rightarrow e'$) if one of the following is true:
  1. $\exists\ p_i \mid e \rightarrow_i e'$
  2. $e = send(m) \wedge e' = receive(m)$

     e is the event of sending message *m*, and *e'* is the corresponding receive event
  3. $\exists\ e, e', e'' \mid (e \rightarrow e'') \wedge (e'' \rightarrow e')$

     The happened-before relation is transitive
- By applying these rules, we can construct a sequence of events $e_1, e_2, \ldots, e_n$ that are causally ordered
- Observations
  - The happened-before relation defines a partial ordering (properties: *non-reflexive, anti-symmetric*, *transitive*)
  - The sequence $e_1, e_2, \ldots, e_n$ is not necessarily unique
  - For a pair of events, they are not always related by a happened-before relation. In such cases, the events are said to be concurrent (denoted by ||)

Happened-before relation: example



- Sequence $s_1 = e_1^1, e_2^1, e_2^2, e_3^2, e_3^3, e_1^3, e_1^4, e_1^5, e_2^4$
- Sequence $s_2 = e_3^1, e_1^2, e_1^3, e_1^4, e_3^5$
- Can you find another sequence?

- Events e_3^1 and e_2^1 are concurrent

  $e_3^1 \not\rightarrow e_2^1$ and $e_2^1 \not\rightarrow e_3^1$

Se due eventi avvengono in due processi diversi che non scambiano messaggi, tali eventi sono detti concorrenti. Ciò significa che semplicemente non si può dire nulla sul quando sono accaduti e l'ordine

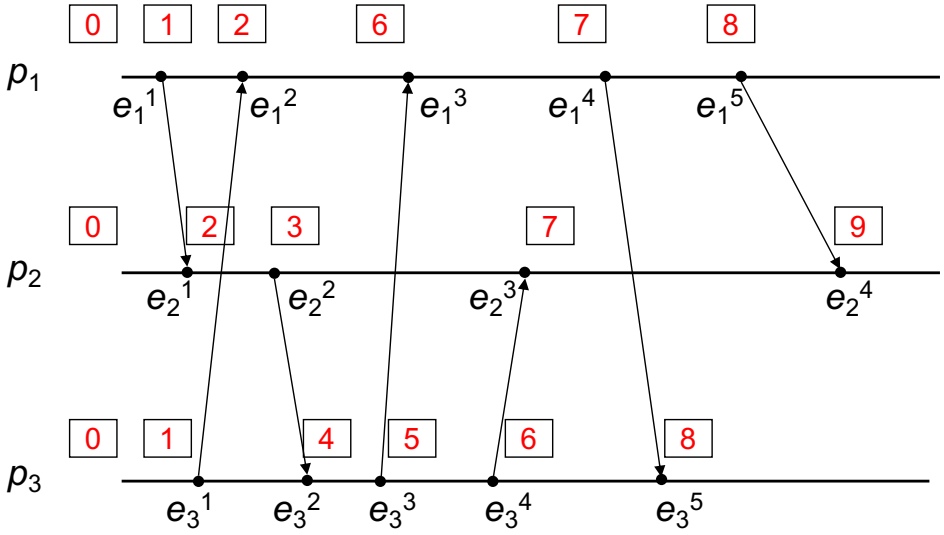
Scalar logical clock

- A monotonically increasing software counter (scalar value), used to order events in DS
  - Scalar logical clock (or Lamport clock): we call it "scalar" to distinguish from vector clock
- Does not rely on physical clock
- Each process p_i has its own scalar clock L_i, and it uses it to apply timestamps to events
- Denote $L_i(e)$ as the timestamp, based on the scalar clock, applied by process p_i to event e
- Key property: **if $e \rightarrow e'$ then $L(e) < L(e')$**
- Key observations:
  - If $L(e) < L(e')$, it does not necessarily mean that $e \rightarrow e'$; however, $L(e) < L(e')$ implies that $e \rightarrow e'$
  - The happened-before relation introduces a partial ordering of events: in the case of concurrent events, it is impossible to determine which event actually happens first

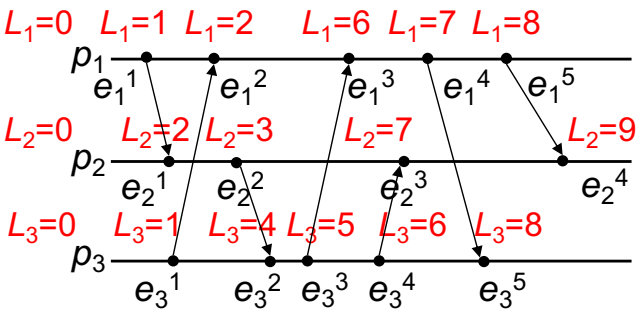
Scalar clock: update

- Scalar clock update algorithm by Lamport
- Initialization: each process p_i initializes its scalar clock L_i to 0 ($\forall\ i = 1, \ldots, N$)
- Internal event: before executing an internal event, p_i increments L_i by 1: $L_i = L_i + 1$
- Sending a message: when p_i sends message m to p_j
  - Increment the value of L_i: $L_i = L_i + 1$
  - Attach the timestamp $t = L_i$ to message m
  - Execute the *send*(m) event
- Receiving a message: when p_j receive message m with timestamp t
  - Update its scalar clock $L_j = max(t, L_j)$
  - Increment the value of L_j: $L_j = L_j + 1$
  - Execute the *receive*(m) event

Scalar clock: example



Scalar clock: example

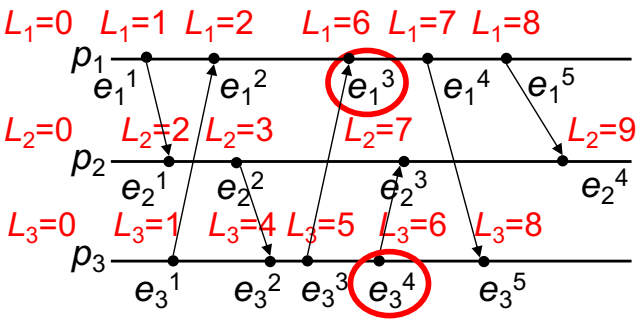


- Observations
  - $e_1^1 \rightarrow e_2^1$ and the respective timestamps reflect this causal relationship (L_1=1 and L_2=2); indeed, *if* $e_1^1 \rightarrow e_2^1$ *then* $L(e_1^1) < L(e_2^1)$
  - $e_1^1 \parallel e_3^1$ and the respective timestamps are equal (L_1=1 and L_3=1); indeed, *if* $L(e_1^1) \geq L(e_3^1)$ then $e_1^1 \not\rightarrow e_3^1$
  - $e_2^1 \parallel e_3^1$ and the respective timestamps differ (L_2=2 and L_3=1); indeed, *if* $L(e_2^1) \geq L(e_3^1)$ then $e_2^1 \not\rightarrow e_3^1$

Total ordering of events

- Problem: using a scalar clock, two or more events can have the same timestamp. How can we establish a total ordering of events to avoid two events occurring at the same logical time?
- Solution: in addition to the scalar clock, we use the process number on which the event occurred
  - We establish a total ordering ($\prec$) between the processes
- Total order relation between events, denoted by $e \Rightarrow e'$: if e is an event on process p_i and e' is an event on process p_j, then $e \Rightarrow e'$ if and only if:
  1. $L_i(e) < L_j(e')$ esempio: <40,2> e <40,3> allora si prende l'evento del processo con id 2. <39,2> e <38,3> si prende l'evento del processo con id 3.
  2. $L_i(e) = L_j(e') \wedge p_i \prec p_j$
- This relation is used in some distributed mutual exclusion algorithms to ensure a total ordering of events across processes

Total ordering of events: example



- e_1^3 and e_3^4 have the same scalar clock value: how to order them?
- $e_1^3 \Rightarrow e_3^4$ since $L_1(e_1^3) = L_3(e_3^4)$ and $p_1 \prec p_3$

Scalar clock and its limitation

- The scalar clock has the following property
  - If $e \rightarrow e'$, then $L(e) < L(e')$
- However, it is not possible to guarantee that:
  - If $L(e) < L(e')$, then $e \rightarrow e'$
    See slide 36: $L(e_3^1) < L(e_2^1)$ but $e_3^1 \,||\, e_2^1$
- Consequence: it is not possible to determine, by just looking at scalar clocks, whether two events are concurrent or not
- How to overcome this limitation?
- To solve this issue, vector clocks were introduced by Mattern (1989) and Fidge (1991)

Vector clock

- A vector clock for a system with N processes is a vector of N integers
- Each process p_i maintains its own vector clock V_i vettore di vettori
- For process p_i, $V_i[\, i \,]$ is the local scalar clock
- Each process uses its own vector clock to assign timestamps to events
- Similar to Lamport's clock, the vector clock is attached to the message m, and the timestamp becomes vector-valued
- The vector clock captures the full characteristics of the happened-before relation

$$e \rightarrow e' \text{ if and only if } V(e) < V(e')$$

Vector clock: meaning and comparison

- Given the vector clock V_i
  - $V_i[\,i\,]$ is the number of events generated by p_i
  - $V_i[\,j\,]$ with $i \neq j$ is the number of events that occurred at p_j, which p_i knows about

- Comparison of vector clocks
  - $V = V'$ if and only if $\forall\, j$: $V[\,j\,] = V'[\,j\,]$
  - $V \leq V'$ if and only if $\forall\, j$: $V[\,j\,] \leq V'[\,j\,]$
  - $V < V'$ (and thus the event associated with V *precedes* the event associated with V') if and only if
    - $\forall\, i \in [1, \ldots, N]$: $V[\,i\,] \leq V'[\,i\,]$

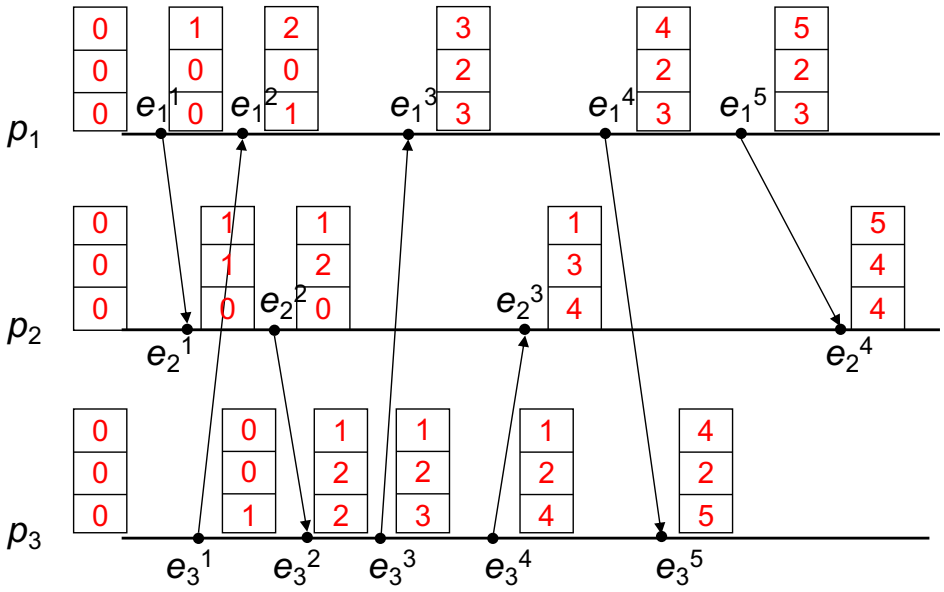
      and
    - $\exists\, j \in$ in $[1, \ldots, N]$: $V[\,j\,] < V'[\,j\,]$
  - $V \,||\, V'$ (and thus the event associated with V *is concurrent* to the event associated with V') se e solo se
    - $\neg(V < V')$ and $\neg(V' < V)$

Vector clock: update

- Initialization: each process p_i initializes its own vector clock V_i: $V_i[k]=0 \;\; \forall\, k = 1, 2, \ldots, N$
- Internal event: before executing an internal event, p_i increments its own vector clock component $V_i[i]$ by 1: $V_i[i] = V_i[i] + 1$
- Sending a message: when p_i sends message m to p_j
  - Increment its own clock component $V_i[i]$ by 1: $V_i[i] = V_i[i] + 1$
  - Attach the vector timestamp $t = V_i$ to message m
  - Execute the *send(m)* event
- Receiving a message: when p_j receives message m with timestamp t:
  - Update its vector clock component for each process K: $V_j[k] = max(t[k], V_j[k]) \;\forall\, k = 1, 2, \ldots, N$
  - Increment its own clock component $V_j[j]$ by 1: $V_j[j] = V_j[j] + 1$
  - Execute the *receive*(m) event

non aumenta il valore di tutti i clock, ma solo quello associato a j

Vector clock: example



Using vector clocks to compare events

- Let $V(e)$ and $V(e')$ be the vector timestamps of two events e and e'
- By comparing $V(e)$ and $V(e')$, we can determine if the events are in happened-before relationship or concurrent

V	V'
1	1
2	2
0	2

$V(e) < V'(e)$ and thus $e \rightarrow e'$

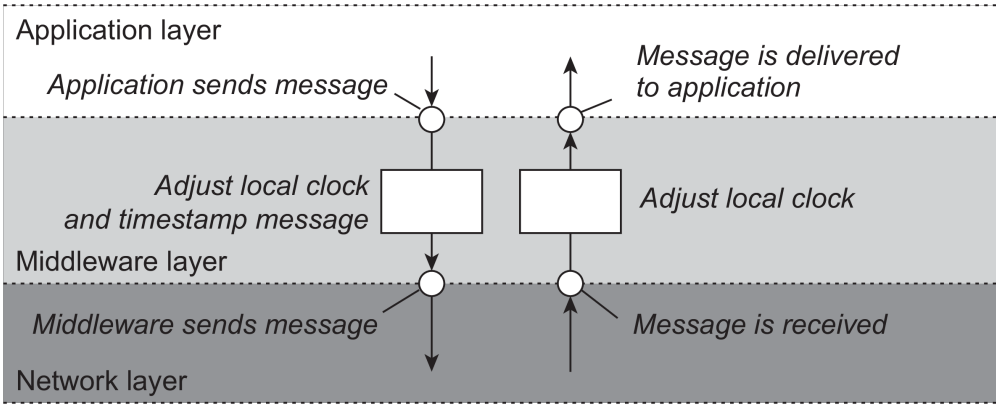
V	V'
1	1
2	0
0	2

$V(e) \neq V'(e)$ and thus $e \parallel e'$
nei componenti 2 e 3 sono diversi!

Examples of applications of logical time

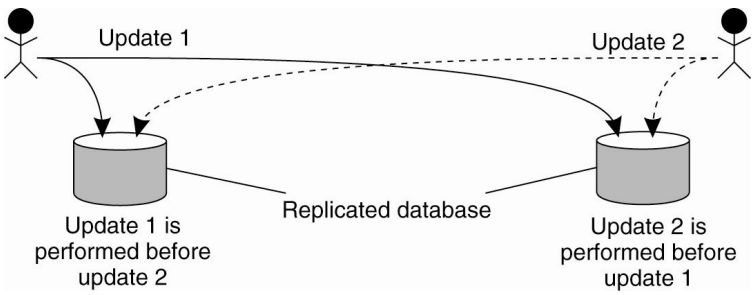
- We will examine two applications of logical time:

1. Scalar clock for totally ordered multicasting
2. Vector clock for causally ordered multicasting



Totally ordered multicasting: problem

- To ensure that concurrent updates on a replicated database are seen in the same order by every replica, you need to synchronize and order the updates across all replicas, ensuring consistency
  - p_1: add \$100 to a bank account (initial value: \$1000)
  - p_2: increment the account by 1%
  - There are two replicas and, without synchronization, each replica may apply the updates in a different order
    - Replica #1 ← 1111 (first p_1 and then p_2)
    - Replica #2 ← 1110 (first p_2 and then p_1)



Totally ordered multicasting

- A totally ordered multicast is a multicast operation in which all messages are delivered in the same order to every recipient
- Assumptions: abbastanza forti
  - Reliable communication: no message loss occurs
  - FIFO ordered communication: messages sent from p_i to p_j are received by p_j in the same order in which p_i sent them p_j

- Let's consider two solutions to implement totally ordered multicasting
  - Centralized
  - Distributed, using scalar clocks

Totally ordered multicasting: solutions

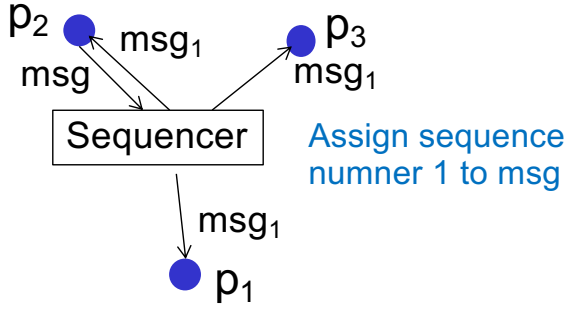
- Centralized solution: a single coordinator (*sequencer*) manages the order of messages)
- The sequencer ensures that messages are delivered to all recipients in the same order
  - Each process p_i sends its update message to the sequencer
  - The sequencer then assigns a unique sequence number to each update message and multicasts the message
  - Each process p_i executes the update in the order of the sequence numbers

✓ Simple to implement

✗ Single point of failure

✗ Potential bottleneck at the sequencer

ovviamente non è la soluzione ideale



Totally ordered multicasting: solutions

- Distributed algorithm using scalar clocks
  - Tag message with scalar clock: each update message msg_i sent by process p_i is tagged with p_i's scalar clock at the time of sending
  - Multicast the message: p_i multicasts to all processes (including itself)
  - Queue messages locally: each receiving process p_j places msg_i in a local queue $queue_j$, sorted by timestamp (scalar clock value) questo eventualmente porta tutti ad avere la stessa coda ordinata
  - Acknowledge receipt: p_j multicasts an ack for msg_i to all processes, indicating it has received the message
  - Deliver message to application: p_j delivers msg_i to the application only if all of the following conditions are met:

attenzione a non confondersi, gli acks devono arrivare agli altri processi

1. msg_i is at the head of $queue_j$ (and all acks for msg_i have been received by p_j)
2. for every other process p_k, there is a message msg_k in $queue_j$ with a timestamp greater than msg_i ribadisce che msg_i debba essere in testa alla coda

msg_i is delivered only when p_j knows that no other process can multicast a message with a smaller or equal timestamp than msg_i: this ensures that all processes deliver messages in the same total order

Quando viene affidato all'applicazione il messaggio è rimosso dalla coda e gli ack rimossi

La soluzione funziona perché ogni processo ha la stessa copia della coda, quindi tutti i messaggi sono consegnati con lo stesso ordine ovunque.

Totally ordered multicasting: distributed

✓ Decentralized: no single point of failure and works better than centralized one with larger systems

✗ Slightly more complex than centralized solution

✗ Communication cost:

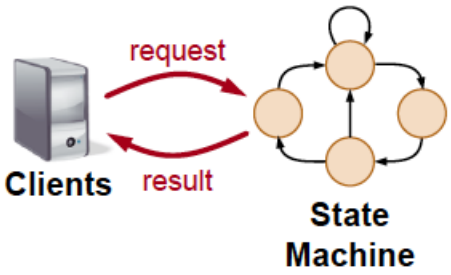
- Ack overhead: $O(N^2)$ acks for each message
  - Each process sends N-1 acks for each message it receives, and this happens for every process in the system

inoltre se c'è molta latenza oppure i processi falliscono la soluzione non è perfetta

- Mechanism designed by Lamport for **state-machine replication**

State machine replication

- Software replication technique applied to a service that can be implemented as a deterministic state machine
  - Used to replicate the state of a machine ("state" can be any data or process state) across multiple replicas
  - A state machine consists of a series of states and transitions between those states
  - Each operation in the system causes a transition between states: next state of service = f(current state, op)
  - Deterministic state machine: for every given state and operation, there is exactly one defined next state
- Each replica processes the same sequence of operations in the same order, ensuring that all replicas stay in sync (i.e., end up in the same state)



State machine replication

- To achieve fault tolerance, service is replicated on several nodes, all of them running a state machine replication (SMR) middleware
  - Set of replicas behaves as a "centralized" server
- Service makes progress as long as any majority of replicas are up
- Replicas can recover by reapplying the operations in the same order
- Used in systems that require strong consistency and fault tolerance, such as distributed databases, file systems, and distributed transaction systems

State machine replication

- Lamport's totally ordered multicast ensures consistency, but it may face scalability challenges
  - Particularly in systems with high latency or frequent failures appunto
- Other well-known solutions for ensuring state-machine replication we examine
  - Paxos consensus algorithm
  - Raft consensus algorithm

che vedremo entrambi

Causally ordered multicasting

- Ensures that a message is delivered only after all the messages that causally precede it (i.e., events that are causally related) have already been delivered
  - Causal relation between two events: the second event is *potentially* influenced by the first event, meaning there is a cause-effect relationship
  - Goal: deliver the cause before the effect to maintain the causal order of events
  - Weaker than totally ordered multicasting
    - We do not require all messages to be delivered in a global total order. Instead, we only require that causal relationships are respected
  - Example:
    - p_1 sends messages m_A and m_B
    - p_2 sends messages m_C and m_D
    - m_A causes m_C
    - Delivery sequences that are compatible with causal ordering (and FIFO ordering): $m_A\, m_B\, m_C\, m_D$ $m_A\, m_C\, m_B\, m_D$ $m_A\, m_C\, m_D\, m_B$ but $m_C\, m_A\, m_B\, m_D$ is not valid

come per la teoria del clock vettoriale, l'esempio fa capire meglio che le parole

Causally ordered multicasting

anche qui sono forti

- Assumptions: reliable and FIFO ordered communication
- Use vector clocks to solve the problem of causally ordered multicasting in a decentralized way
- Algorithm
  - Sending the message: process p_i sends the message m to all other processes, attaching a vector timestamp (ts_m). The timestamp is p_i's vector clock (V_i) at the time of sending the message
  - Receiving the message: *w*hen p_j receives m from p_i, p_j does not immediately deliver the message to the application. Instead, p_j places the message in a waiting queue
    - $V_j[i]$ counts the number of messages sent by p_i to p_j
  - Conditions for delivering the message: m will be delivered to the application only if both of the following conditions are met:
    1. $ts_m[i] = V_j[i] + 1$ — p_j has received the next message it expects from p_i
    2. $ts_m[k] \leq V_j[k] \ \forall \ k \neq i$ — p_j has seen at least all the messages that p_i has seen up to that point

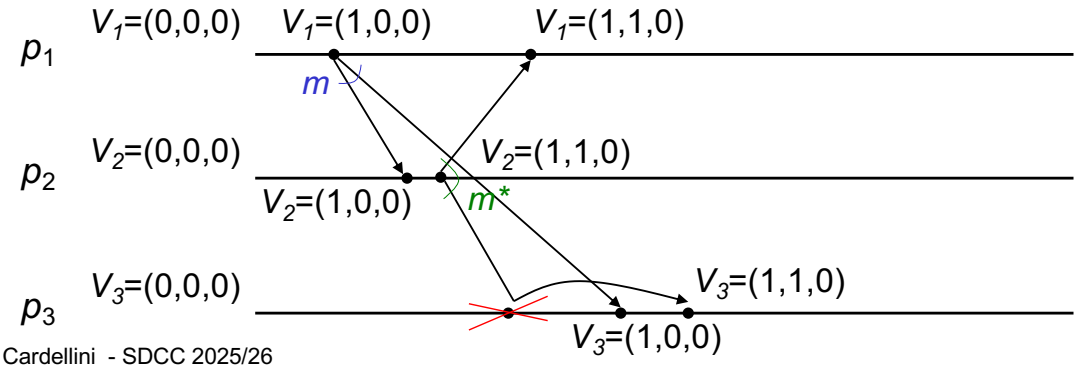
Causally ordered multicasting: example

- p_1 sends m to p_2 and p_3
- p_2 *receives* m and then sends m^* *to* p_1 and p_3
  - Cause-effect relationship between m and m^*
- Suppose that p_3 receives m^* before it receives m
  - The algorithm avoids the violation of causal ordering: p_3 will hold m^* in a waiting queue until m is delivered first

Clock update

p_i sends msg: $V_i[i] = V_i[i] + 1$

p_j receives msg with ts_{msg}: $V_j[k] = \max\{ V_j[k], ts_{msg}[k] \}$



viene rifiutato perché il clock vettoriale di p2 è (1,1,0) mentre quello di p3 è (0,0,0) quindi siccome ho che il messaggio di p1 non è stato ricevuto ovvero V3=(1,0,0) metto in coda il messaggio di p2!

Timestamps in practice

- Timestamps are useful for comparing events in DS, e.g.
  - Reconciling object updates in distributed storage system
  - Restoring system state after crash

    1. Checkpoint the system: a checkpoint is created, capturing the system's state at a specific point in time; 2 Record events: each event is recorded with a timestamp to track the order in which events occurred; 3 After a crash: the checkpoint is restored, and the system replays the events in order using the timestamps
- How to compare timestamps across different processes?

1. Physical timestamps: rely on synchronization of physical clocks across processes
   - Use case: Google Spanner uses TrueTime to maintain consistency
   - Limitation: require all clocks to be synchronized, which can be difficult in geographically distributed systems with high latencies quindi Spanner è utilizzato per sincronizzazione interna non esterna

Timestamps in practice

2. Logical timestamps: use scalar clocks to order events
   - Use case: Oracle DB use systems change numbers based on scalar clocks to track changes to data, ensuring that operations are applied in the correct order
   - Limitation: scalar clocks do not fully distinguish between causally related events and concurrent events che servono magari per sistemi critici con transazioni
3. Vector timestamps: extend logical timestamps by using vector clocks
   - Use case: Amazon DynamoDB uses vector clocks to determine which object version is the most recent by comparing vector timestamps